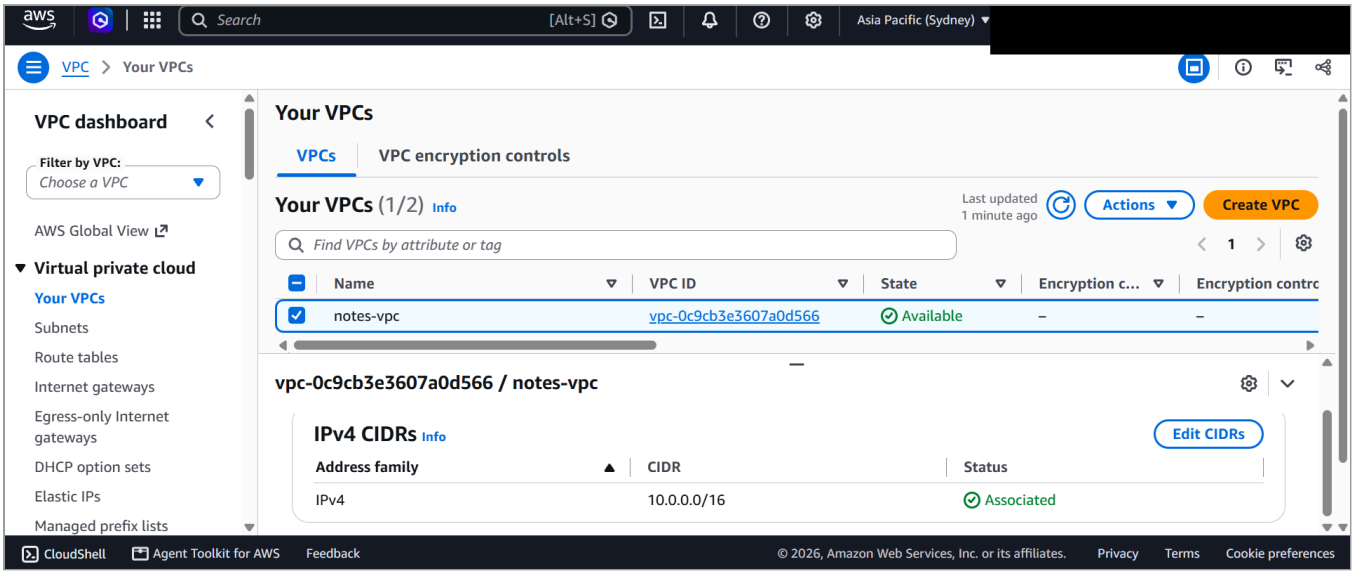
Evidence appendix

NoteNest on AWS (CEA SE-490). AWS console screenshots taken on 5 Nov 2026, region ap-southeast-2. Account identifiers are blacked out. Command-line evidence is in docs/evidence/*.txt.

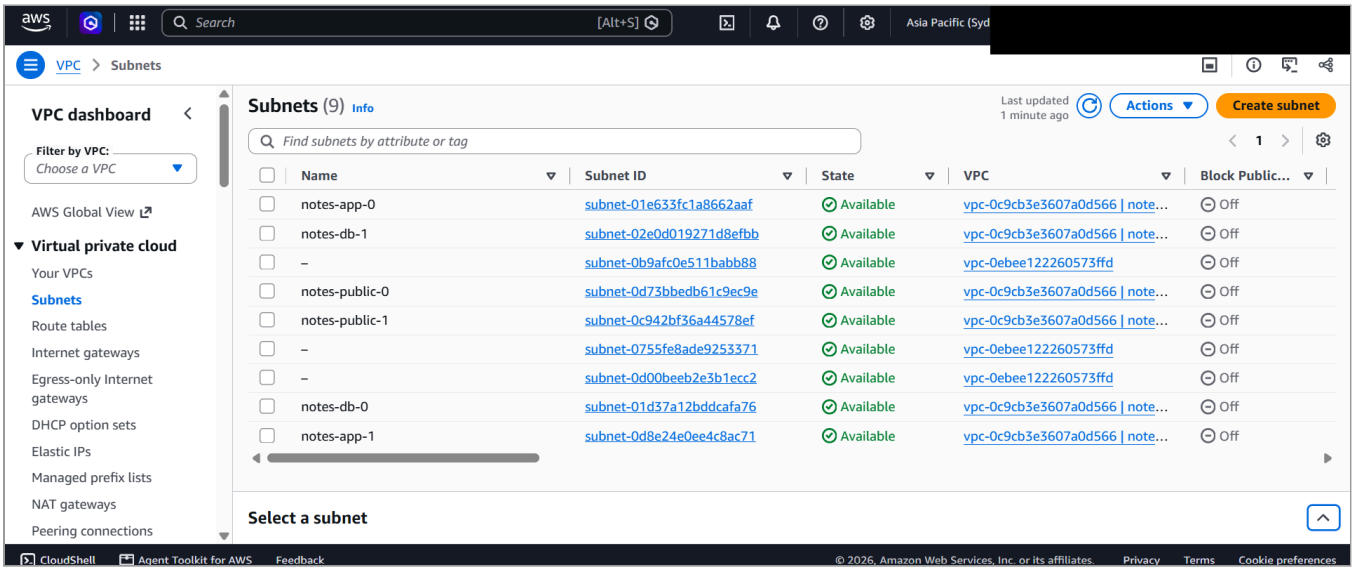
1. VPC

VPC **notes-vpc**, state Available, IPv4 CIDR **10.0.0.0/16** (brief item 1: VPC with CIDR).



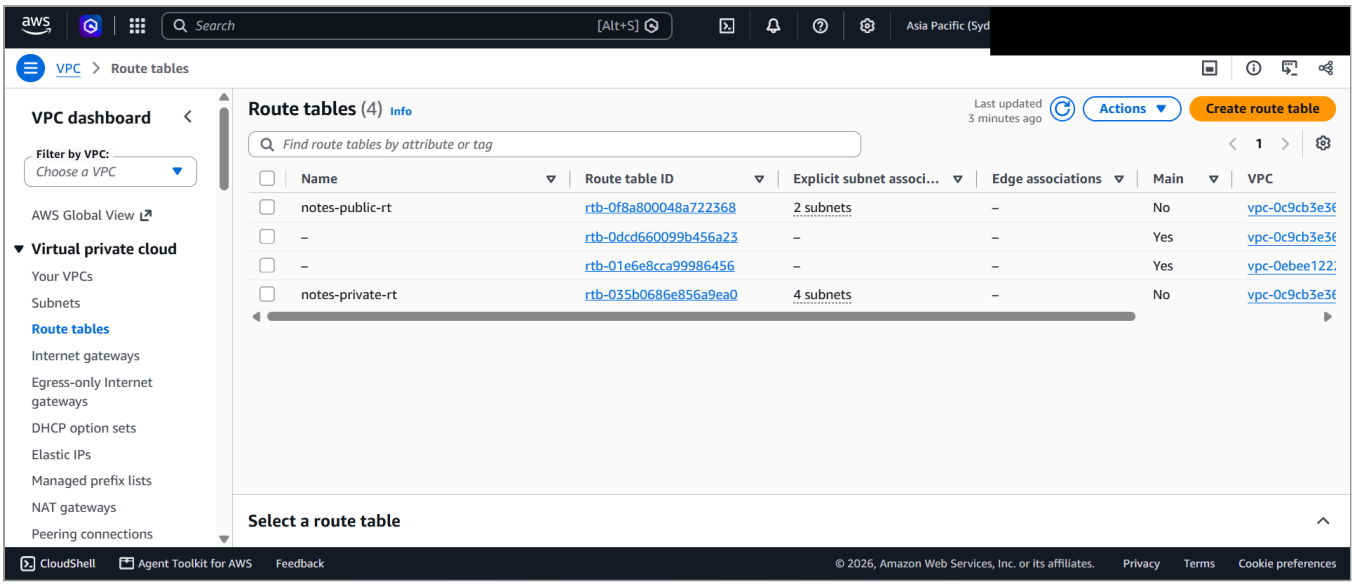
2. Subnets

The project's subnets (notes-public-0/1, notes-app-0/1, notes-db-0/1) across two Availability Zones; the unnamed rows are the account's default-VPC subnets. 'Block public access' is Off at subnet level because exposure is controlled by route tables and security groups; only the public subnets auto-assign public IPs (see cli-evidence.txt).



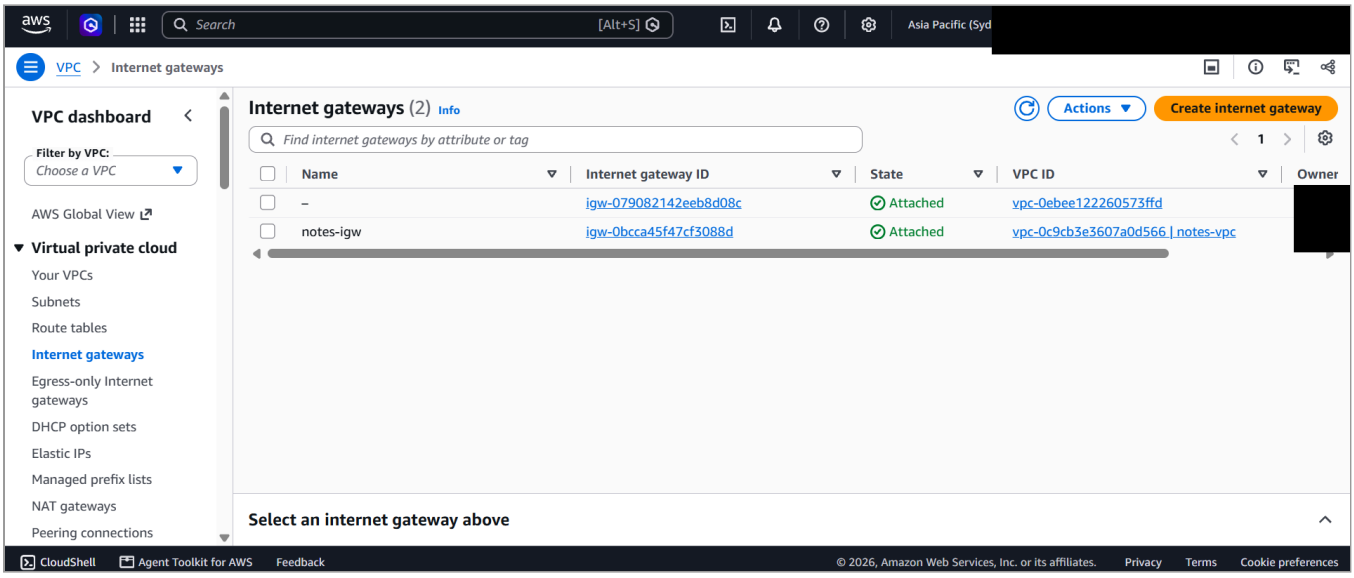
3. Route tables

notes-public-rt (2 subnets) and **notes-private-rt** (4 subnets); the routes themselves (public to Internet Gateway, private to NAT Gateway plus the S3 endpoint) are listed in cli-evidence.txt.



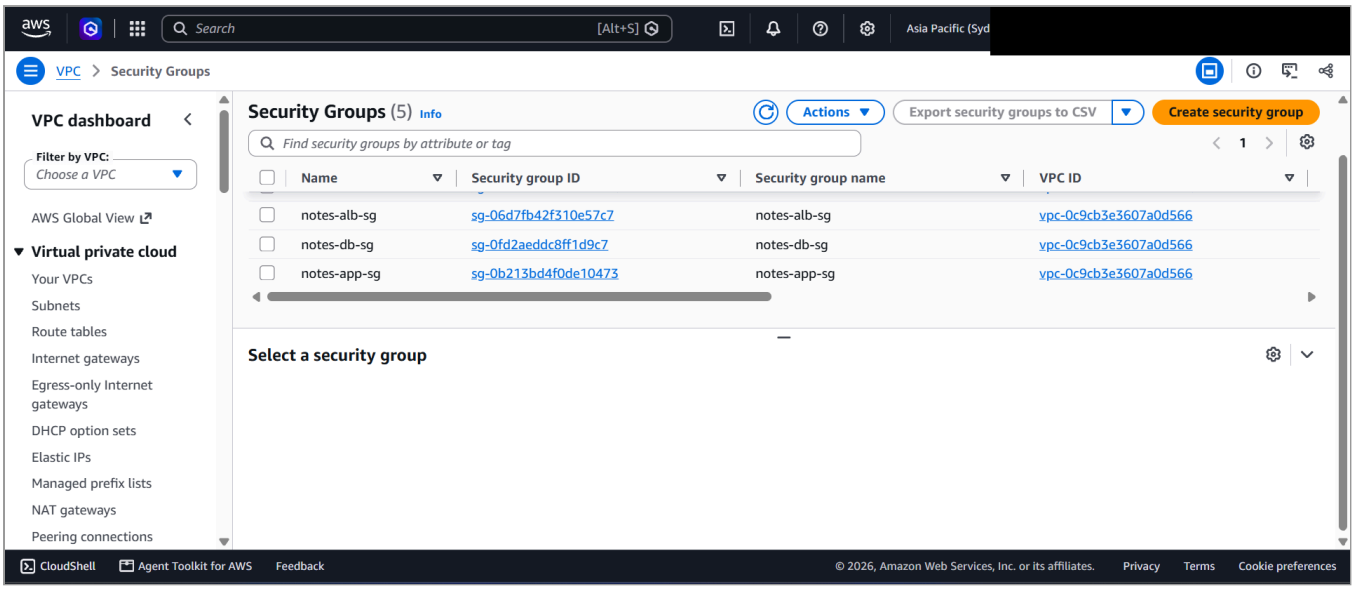
4. Internet gateway

notes-igw attached to the project VPC. The NAT Gateway (state available) and the S3 gateway endpoint are shown in cli-evidence.txt.



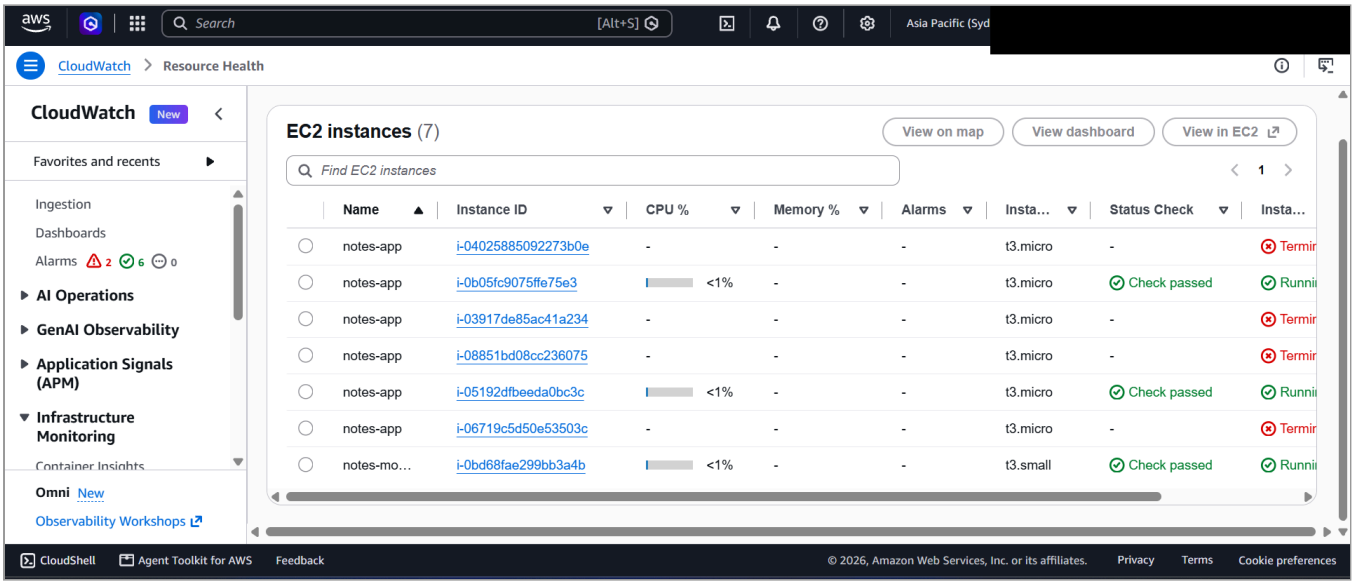
5. Security groups

The three tier security groups **notes-alb-sg**, **notes-app-sg** and **notes-db-sg**. Their rules (SG-to-SG references, no port 22) are listed in cli-evidence.txt.



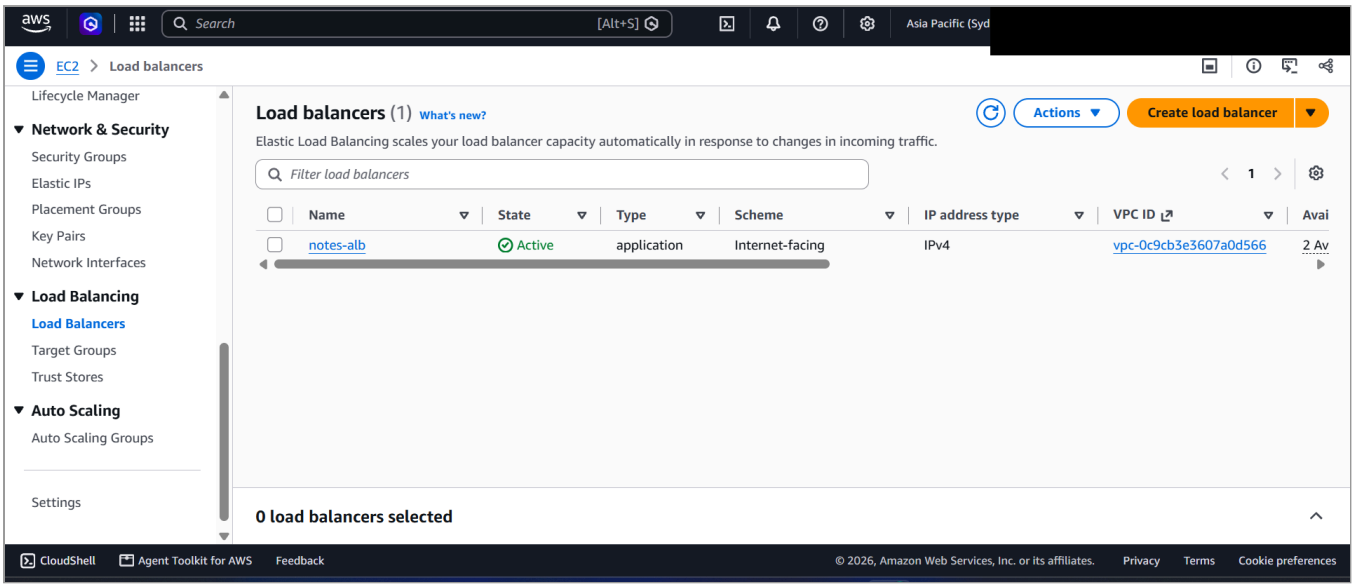
6. EC2 instances (CloudWatch resource health view)

Running notes-app instances and the t3.small notes-mongodb database instance, all passing status checks. The rows marked Terminated are the instances removed during the scale-in, failure and refresh tests.



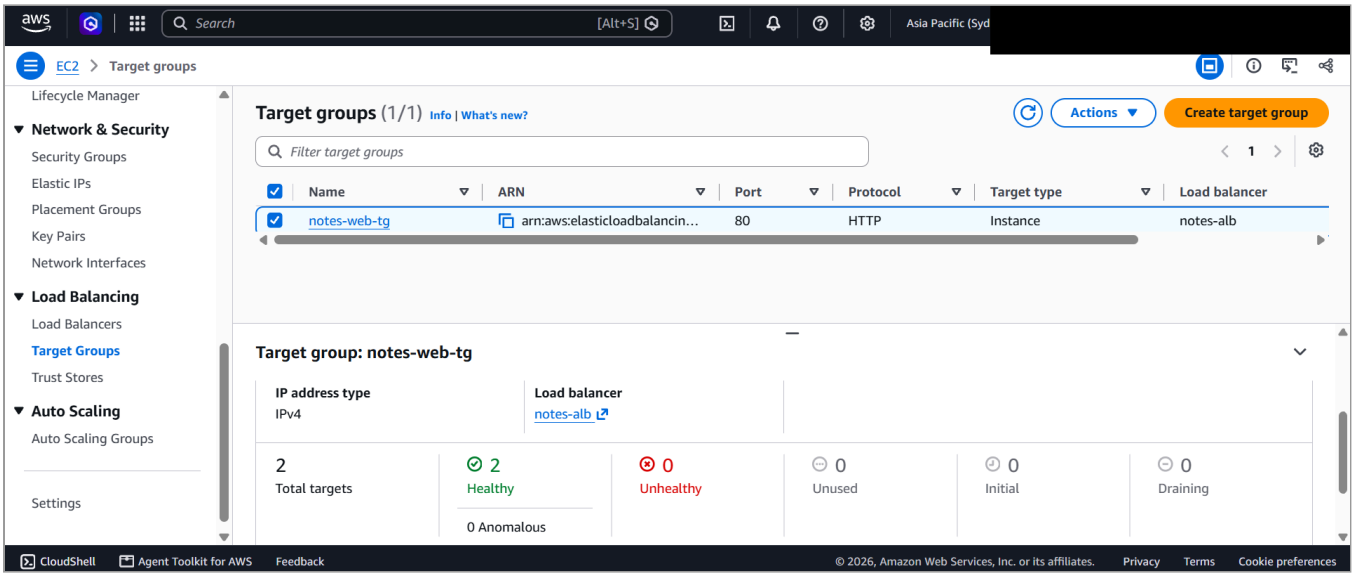
7. Application Load Balancer

notes-alb: type application, Internet-facing, state Active, IPv4, two Availability Zones. DNS name and listener are in cli-evidence.txt and the README.



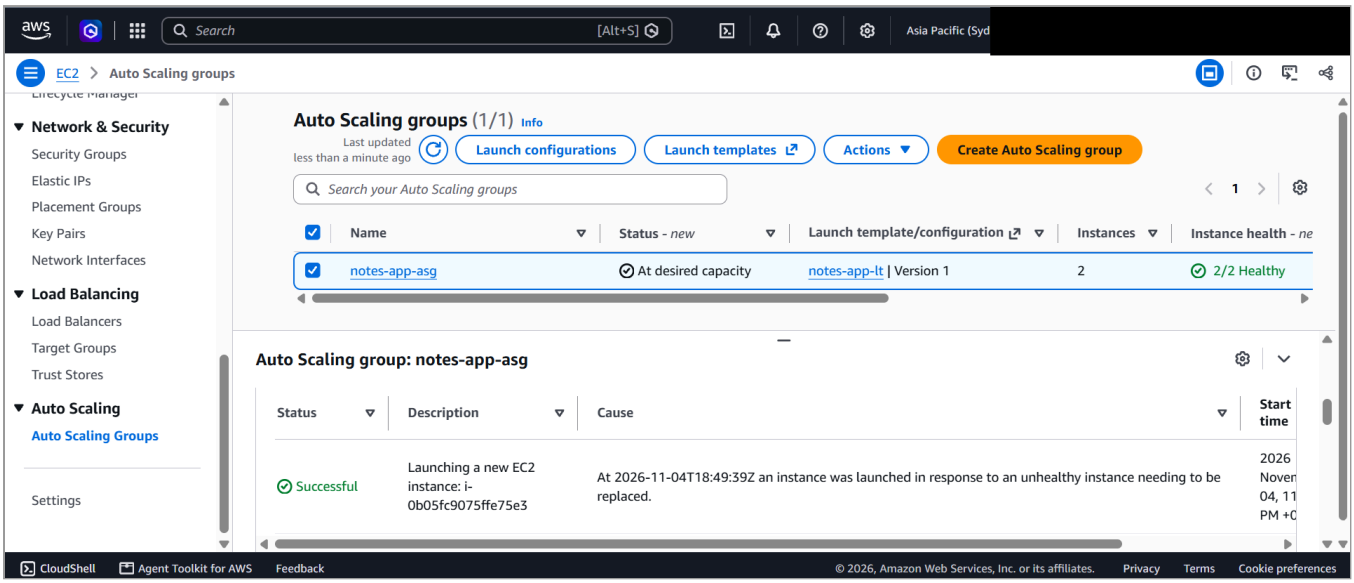
8. Target group

notes-web-tg (HTTP 80, instance targets) attached to notes-alb: 2 total targets, **2 healthy**, 0 unhealthy.



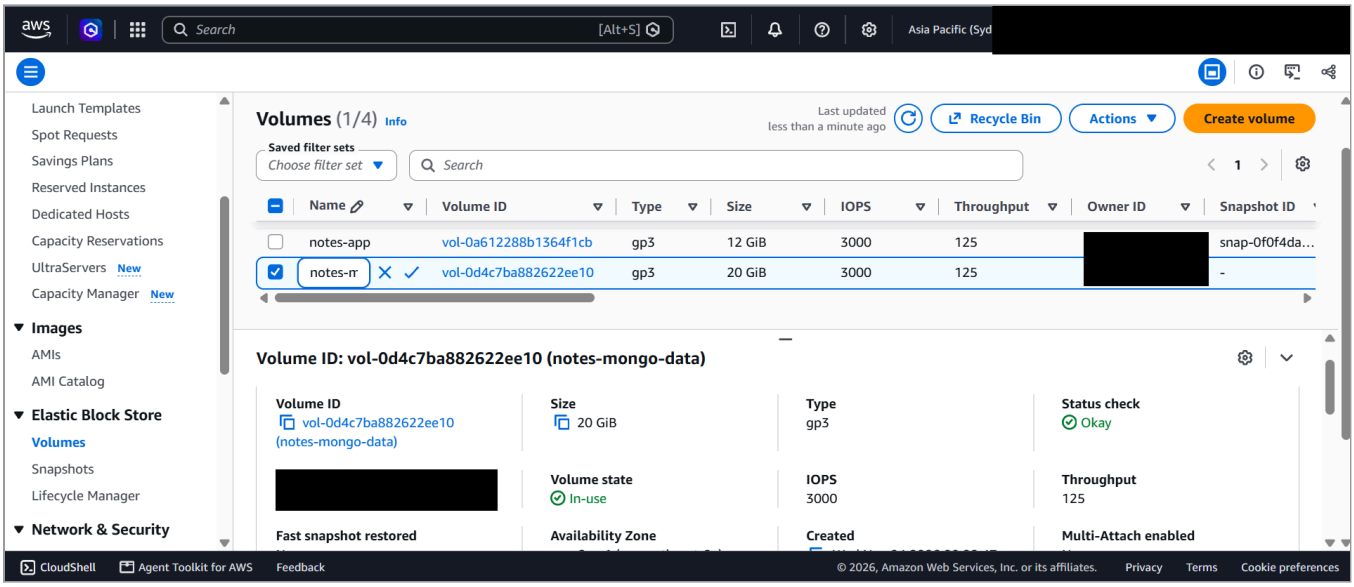
9. Auto Scaling group

notes-app-asg at desired capacity, 2/2 healthy, launch template notes-app-lt, with scaling activity history (a launch to replace an unhealthy instance). Full scale-out and scale-in history is in scaling-timeline.txt.



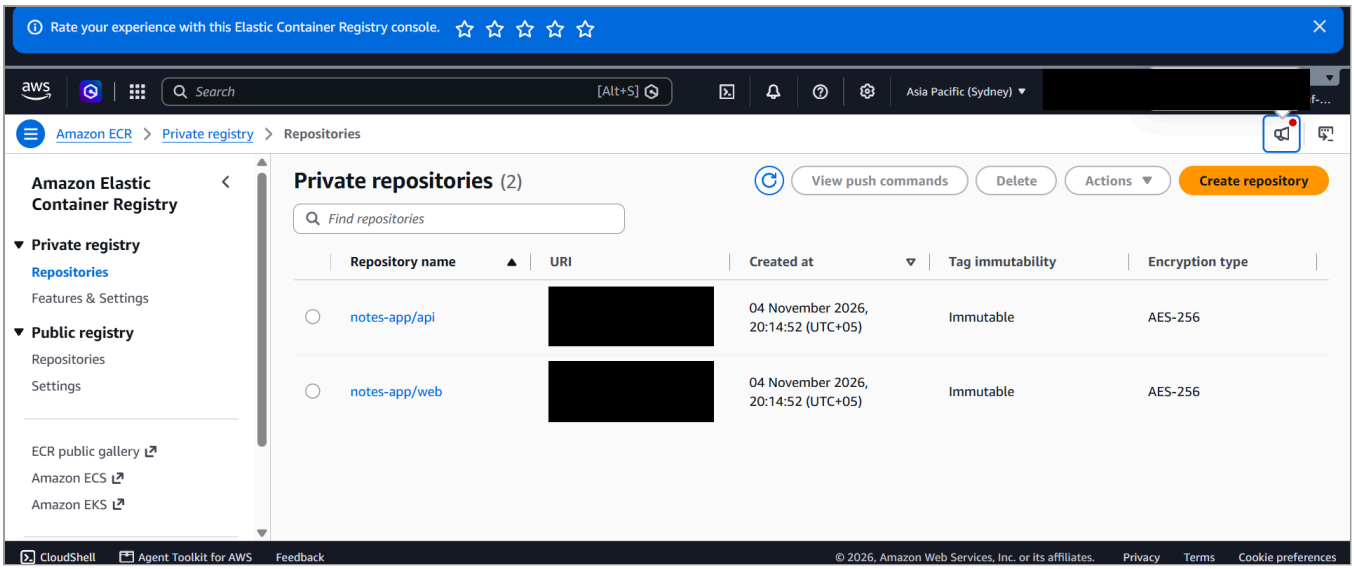
11. EBS data volume

notes-mongo-data: 20 GiB gp3, In-use, status check Okay (encryption confirmed in cli-evidence.txt). The Name field is in edit mode only because the page was clicked.



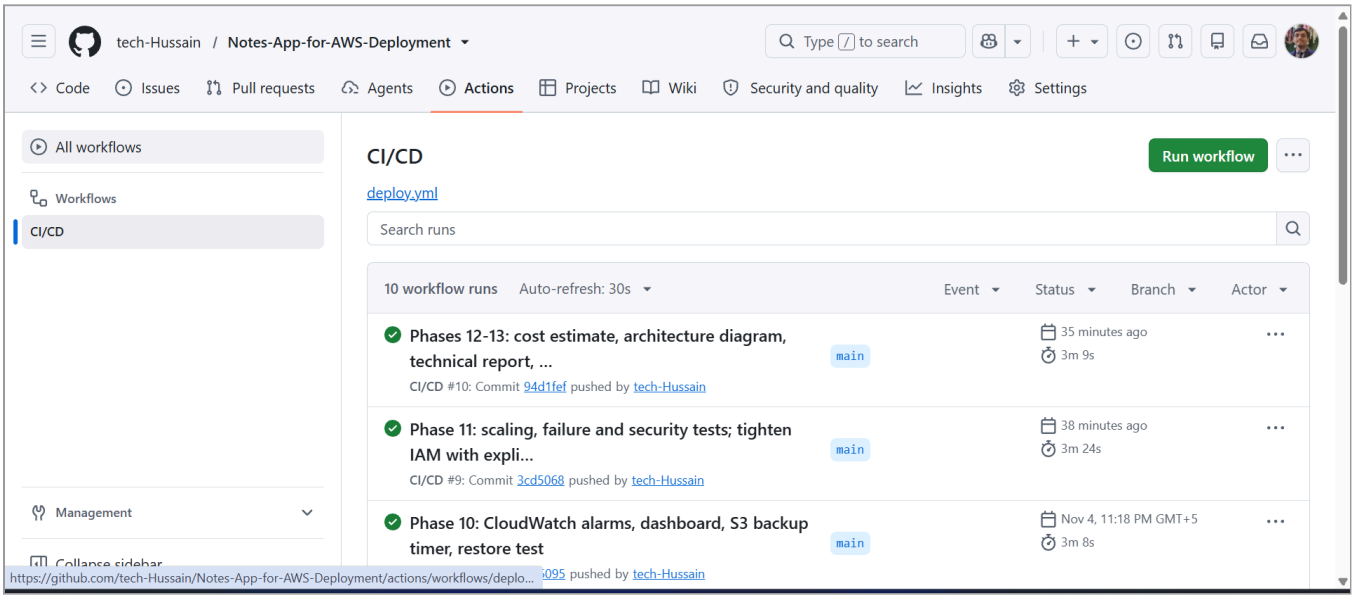
12. Container registry

ECR repositories **notes-app/api** and **notes-app/web** with tag immutability **Immutable** and AES-256 encryption (brief item 5).



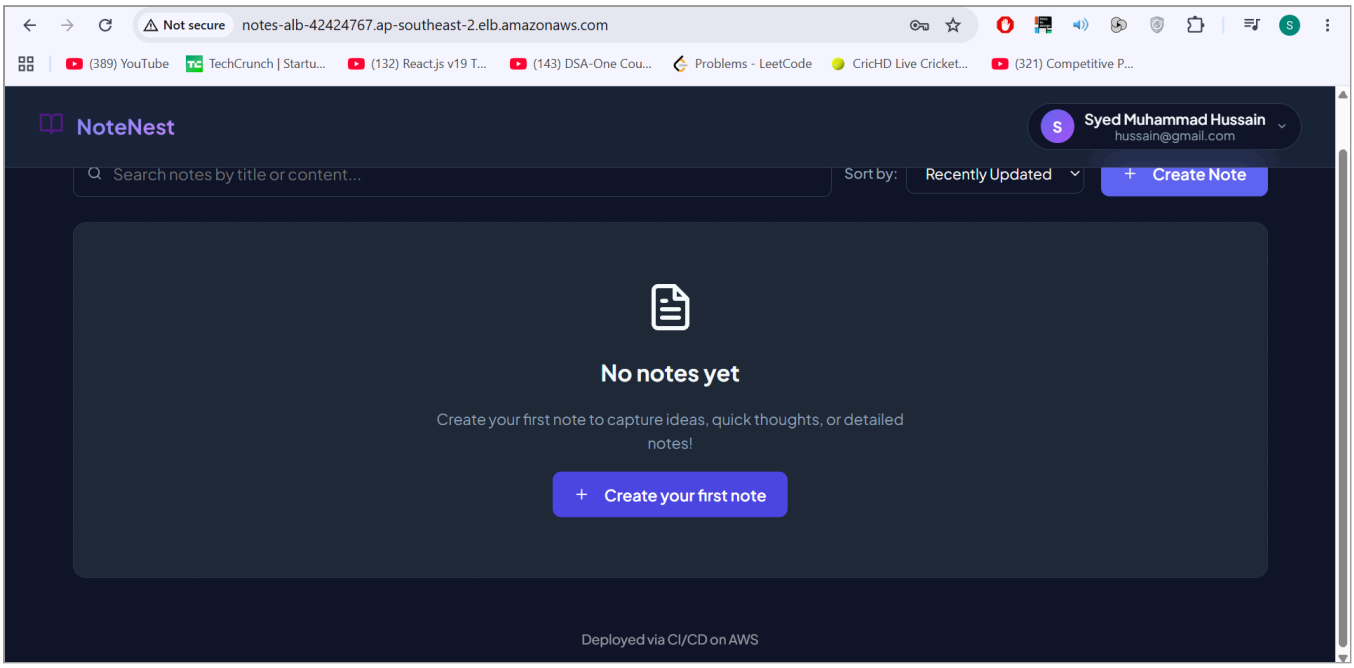
13. CI/CD pipeline runs

GitHub Actions workflow **CI/CD** (deploy.yml): successful runs on main for each commit. Each run executes test, build-push and deploy.



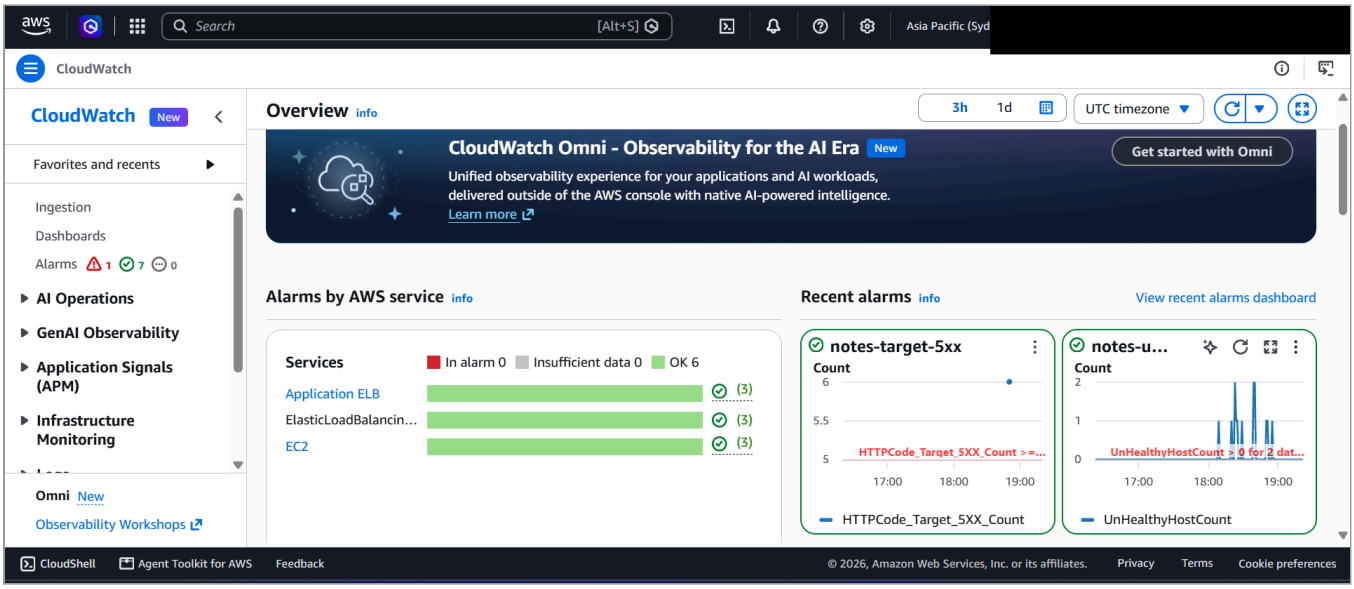
14. Application in the browser

NoteNest served from the load balancer address (**notes-alb-….ap-southeast-2.elb.amazonaws.com**), logged in, with the footer *Deployed via CI/CD on AWS* added by the pipeline test change (brief item 2).



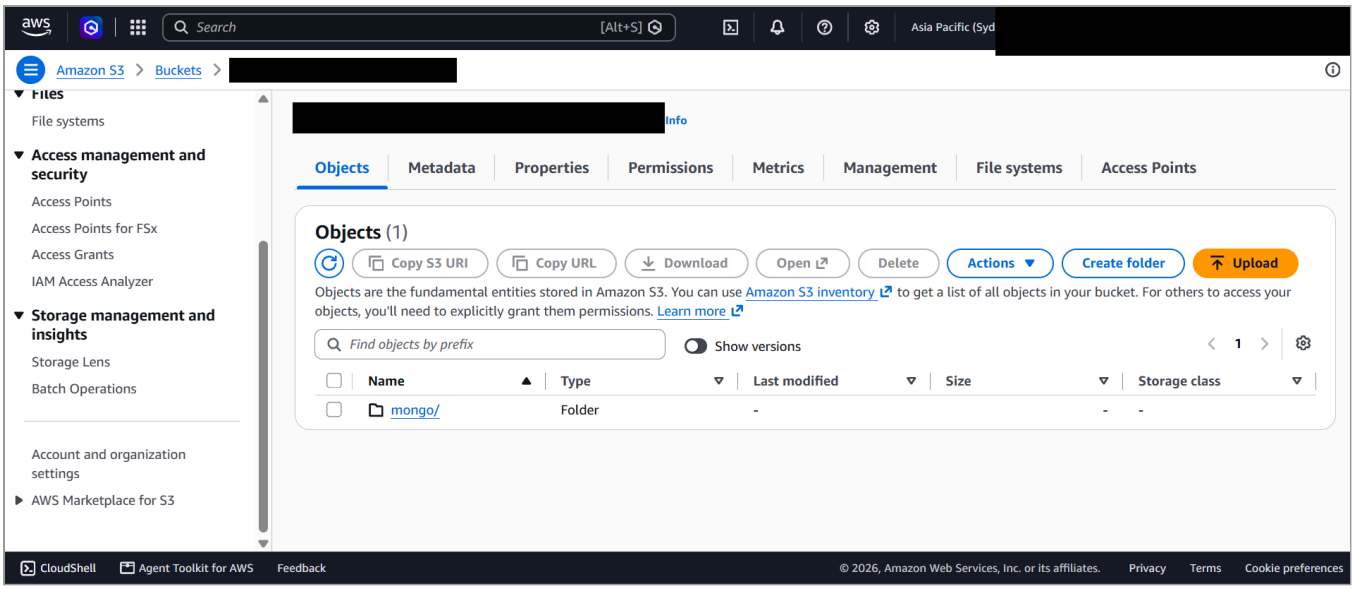
15. CloudWatch overview

Alarms by service: **0 in alarm, 6 OK**, including notes-target-5xx and notes-unhealthy-hosts with their threshold lines (brief enhancement: CloudWatch).



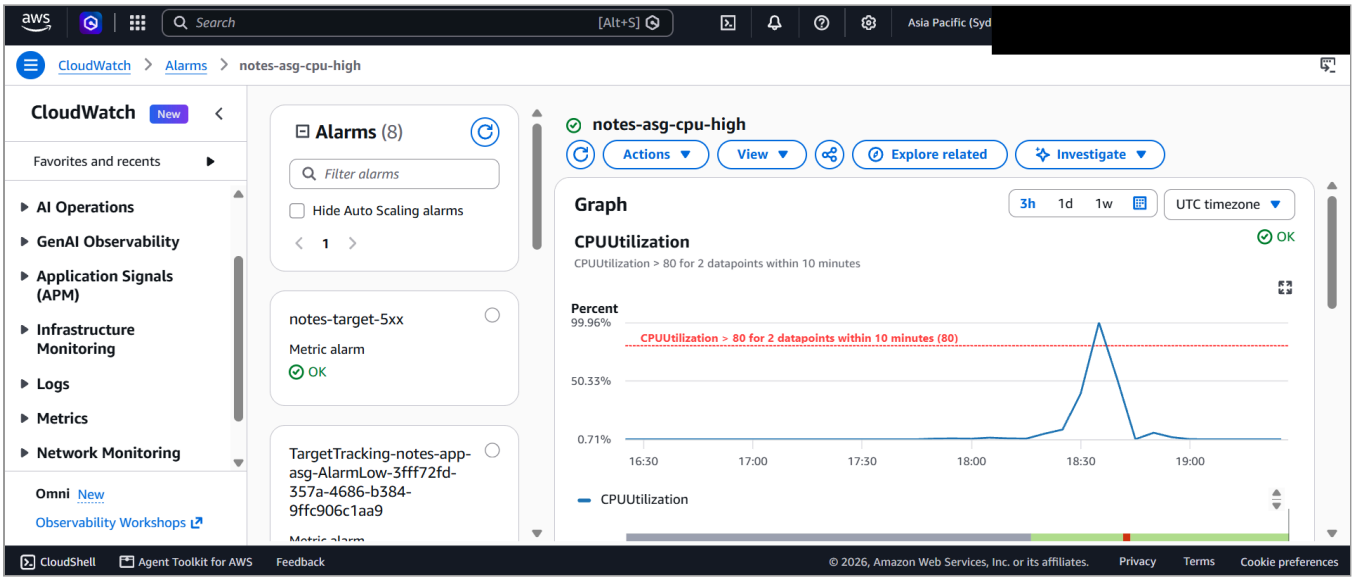
16. S3 backup bucket

The private backup bucket with the **mongo/** prefix holding the nightly MongoDB dumps (brief enhancement: S3 backups). The dump object and the restore test are in backup-restore-test.txt.



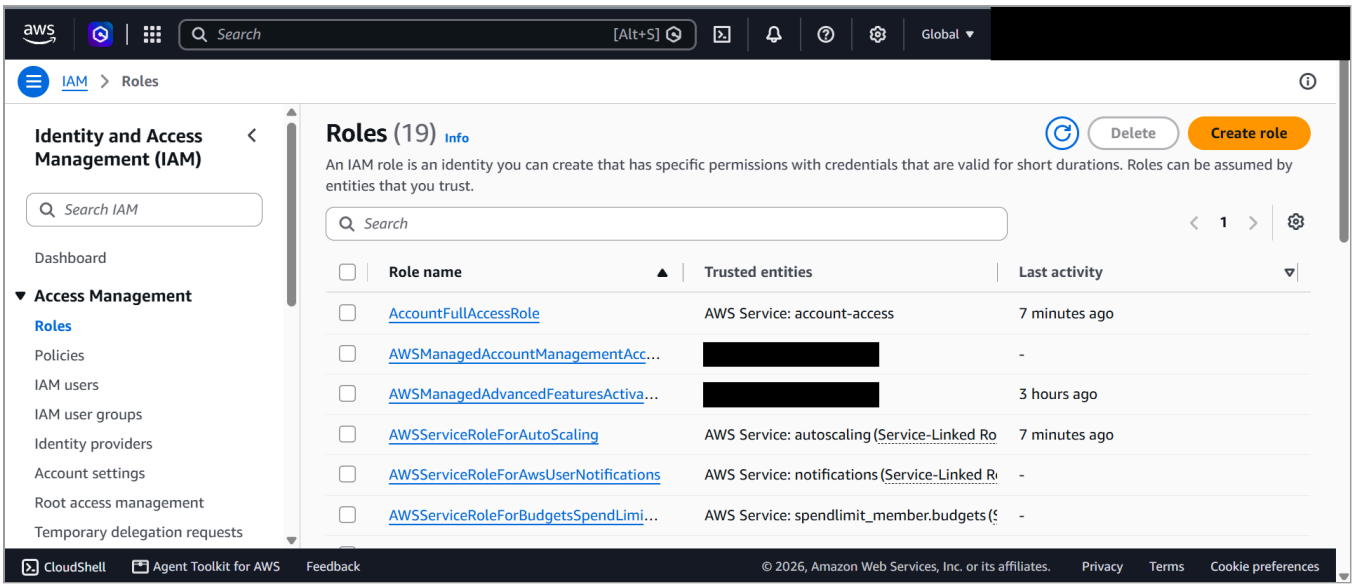
17. Scaling evidence

Alarm **notes-asg-cpu-high**: CPU rising from about 1% to about 100% (the CPU-burn demonstration, around 18:30 UTC) and falling back, matching the 2 to 4 to 2 instance timeline in scaling-timeline.txt.



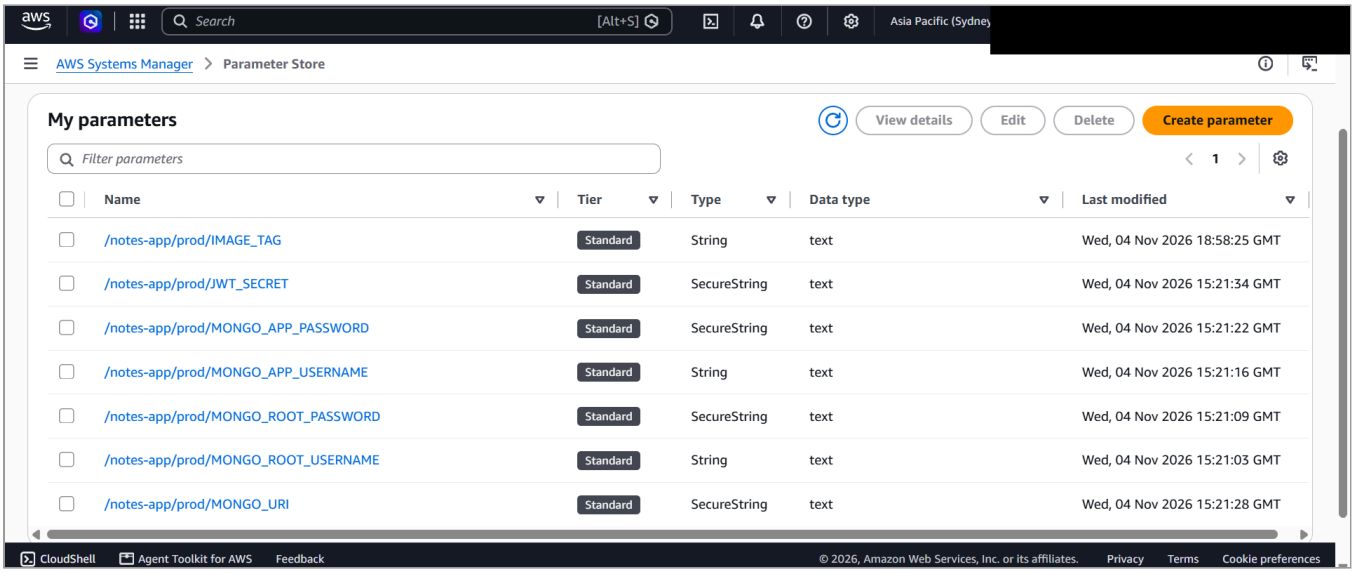
20. IAM roles

IAM role list for the account, including AWSServiceRoleForAutoScaling (used by the scaling group). The project's own roles notes-app-ec2-role, notes-db-ec2-role and notes-github-deploy-role are further down the list; their policies are in infra/iam.tf and infra/github_oidc.tf.



21. Secrets in Parameter Store

Seven parameters under **/notes-app/prod**; secrets (JWT_SECRET, MONGO_URI, passwords) are type **SecureString**. Values are not shown (brief item 8: secrets management).



Extra: app via the load balancer (login page)

Public page served by the load balancer before sign-in.

